Broken Symmetry and Loop Corrections

Positive Mass Case

Consider our φ^4 theory:

$$\mathcal{L} = -\frac{1}{2} Z_\varphi \partial^\mu \varphi \partial_\mu \varphi - \frac{1}{2} Z_m m^2 \varphi^2 - \frac{1}{24} Z_\lambda \lambda \varphi^4$$

Dimensionless for $d=4$ spacetime dimensions.

We are going to look at the case: $m^2 > 0$, Here, the Z_2 symmetry exhibited by the lagrangian is apparent. We have to compute the renormalizing factors.

Using normal dimensional regularization tricks:

$$\Pi(k^2) = \frac{\lambda}{2(4\pi)^2}\left[\frac{2}{\epsilon} + 1 + \ln\left(\frac{\mu^2}{m^2}\right)\right] m^2 - Ak^2 - Bm^2$$

$A = Z_\varphi - 1$ and $B = Z_m - 1$

Here, the loop corrections give us: $A = O(\lambda^2)$; $B = \frac{\lambda}{16\pi^2}\left(\frac{1}{\epsilon} + k_B\right) + O(\lambda^2)$

Arbitrary constant. Zero's out in the $\overline{MS}$ scheme.

Now, looking at the vertex:

$$iV_4(k_1, k_2, k_3, k_4) = -iZ_\lambda \lambda + \frac{1}{2}(-i\lambda)^2 \left(\frac{1}{i}\right)^2 \left[iF(-s) + iF(-t) + iF(-u)\right] + O(\lambda^3)$$

Setting: $Z_\lambda = 1 + c$ gives us:

$$c = \frac{3\lambda}{16\pi^2}\left(\frac{1}{\epsilon} + k_c\right) + O(\lambda^2)$$

Also, the beta function is: $\beta(\lambda) = \frac{3\lambda^2}{16\pi^2} + O(\lambda^3)$

↳ Positive. This means theory gets strongly coupled as energy ↑

Negative Mass Case

Now, we do what we did in the notes that introduced spontaneous symmetry breaking with the shifted fields and stuff:

$$\mathcal{L} = -\frac{1}{2} Z_\varphi \partial^\mu \rho \partial_\mu \rho - \frac{1}{2}\left(\frac{3}{2} Z_\lambda - \frac{1}{2} Z_m\right) m_\rho^2 \rho^2 + \frac{1}{2}(2m - Z_\lambda)\left(\frac{3}{\lambda \tilde{\mu}^\epsilon}\right)^{1/2} m_\rho^3 \rho - \frac{1}{6} Z_\lambda (3\lambda \tilde{\mu}^\epsilon)^{1/2} m_\rho \rho^3 - \frac{1}{24} Z_\lambda \lambda \tilde{\mu}^\epsilon \rho^4$$

We start by looking at the vacuum expectation values: $\langle 0|\rho|0\rangle$.

The three point and one-point vertices contribute to this to give us:

$$\langle 0|\rho(x)|0\rangle = \left(iY + \frac{1}{2}(-iZ_\lambda g_3)\frac{1}{i}\tilde{\Delta}(0)\right)\int d^4y \, \frac{1}{i}\Delta(x-y)$$

Comparing with lagrangian, the factor in the parenthesis is:

$$\frac{i}{2}\left(\frac{3}{\lambda}\right)^{1/2} m_\phi^3\left(2m-2\lambda+\frac{\lambda}{16\pi^2}\left[\frac{2}{\epsilon}+1+\ln\left(\frac{\mu^2}{m_\phi^2}\right)\right]+O(\lambda^2)\right)$$

Using the counter-term definitions this further simplifies to: $\lambda/16\pi^2\left[k_B-k_C+1+\ln(\mu^2/m_\phi^2)\right]$

As is visible, the $1/\epsilon$'s cancel and changing the constants yields a finite result.

For the loop corrections:

$$\Pi(k^2)=-\frac{1}{2}(\lambda\tilde{\mu}^\epsilon)\frac{1}{i}\tilde{\Delta}(0)+\frac{1}{2}g_3^2F(k^2)-x+O(\lambda^2)$$

$$=\frac{\lambda}{32\pi^2}m_\phi^2\left[\frac{2}{\epsilon}+1+\ln\left(\frac{\mu^2}{m_\phi^2}\right)\right]+\frac{3\lambda}{32\pi^2}m_\phi^2\left[\frac{2}{\epsilon}+\int_0^1 dx\ln\left(\frac{\mu^2}{\Delta}\right)\right]-Ak^2-\left(\frac{3}{2}C-\frac{1}{2}B\right)m_\phi^2+O(\lambda^2)$$

Just like this, we find that the divergent terms which are removed in the first case can be removed using the same terms in the second.

This is a general rule for renormalizable theories with spontaneous symmetry breaking.